



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECR 1213

Network Communication

Task 5 – IP Addressing Scheme

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Group Name: TTL Network Solutions

Group members:

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1.0 Introduction

1.1 Task Objectives

The main objective of Task 5 is to design a structured and efficient IP addressing scheme for the new Faculty of Computing building. This task involves subnetting the network address assigned by the lecturer into multiple logical subnetworks that correspond to different labs, classrooms, offices, and wireless access areas. Each subnetwork is allocated based on the number of required hosts to ensure sufficient IP address availability without unnecessary wastage.

Additionally, this task aims to apply appropriate subnetting techniques and logical network segmentation, such as the use of Virtual Local Area Networks (VLANs), to improve network organization and manageability. The final objective is to produce a scalable IP addressing plan that supports current operational needs while allowing for future expansion in line with the faculty's projected growth.

1.2 Importance of IP Addressing

IP addressing plays a critical role in enabling communication between devices within a network. Each device must be assigned a unique IP address to ensure accurate data delivery and prevent address conflicts that could disrupt network operations. In environments with a high number of users and devices, such as academic institutions, improper IP addressing can lead to network instability, performance degradation, and increased troubleshooting efforts.

A well-planned IP addressing scheme enhances network efficiency by reducing broadcast traffic, improving security through logical segmentation, and simplifying network administration. It also enables better scalability, allowing new devices and network segments to be added without major reconfiguration. Therefore, proper IP addressing is essential to maintaining a reliable, secure, and high-performing network infrastructure for the Faculty of Computing.

2.0 Get the Network Address from the lecturer

Group/Section	Network Address
1	172.16.4.0/23
2	172.17.8.0/23
3	172.18.16.0/23
4	172.19.32.0/23
5	172.20.64.0/23
6	172.21.128.0/23
7	172.22.56.0/23
8	172.23.48.0/23
9	172.24.192.0/23
10	172.25.96.0/23

Table 2.0 Network Address from lecturer

Table 2.0 shows the list of network addresses of each group. The network address of our group is 172.16.4.0/23. Our subnet mask is /23 or 255.255.254.0 to ensure that our range of address usage does not influence other group subnets.

Network Address details:

IP address (decimal)	172.16.4.0
IP address (binary)	10101100 00010000 00000100 00000000
Subnet Mask (decimal)	255.255.254.0
Subnet Mask (binary)	11111111 11111111 11111110 00000000

We partitioned the subnet mask into a network portion (23 bits) and a host portion (9 bits). By performing the AND operation of the IP address with the subnet mask, we identified the network address as 172.16.4.0 and the broadcast address as 172.16.5.255. The valid address range within our building spans from 172.16.4.1 to 172.16.5.254.

Network Address	172.16.4.0
Broadcast Address	172.16.5.255
Range of usable Address	172.16.4.1 - 172.16.5.254

3.0 Network distribution for different work areas

Network Address : 172.16. 4. 0/23

Subnet Mask : 1010 1100. 0001 0010. 0000 010 | 0. 0000 0000 / 23

: 1111 1111. 1111 1111.1111 111 | 0. 0000 0000 / 23

(Network Portion)

(Host Portion)

: 255. 255. 254. 0 / 23

i) Determine How Many Bits to Borrow for Each Subnetwork:

Subnetwork	Current Required Host / IP	Reserve for Future Use	Total / IP	Bit Needed
Cisco Networking Lab	32	30	62	$2^x - 2 \geq 62; x = 6$
Embedded Lab	32	30	62	$2^x - 2 \geq 62; x = 6$
General Purpose Lab 1	32	30	62	$2^x - 2 \geq 62; x = 6$
General Purpose Lab 2	32	30	62	$2^x - 2 \geq 62; x = 6$
Core Switch	6	-	6	$2^x - 2 \geq 6; x = 4$
Access Point Ground Floor	6	8	14	$2^x - 2 \geq 14; x = 4$
Access Point First Floor	5	9	14	$2^x - 2 \geq 14; x = 4$
Video Conference Room	2	12	14	$2^x - 2 \geq 14; x = 4$
Hybrid Classroom	5	9	14	$2^x - 2 \geq 14; x = 4$
Lecturer Office	6	8	14	$2^x - 2 \geq 14; x = 4$
Printing Area and Small Meeting Room	3	11	14	$2^x - 2 \geq 14; x = 4$
Router	1	-	1	$2^x - 2 \geq 1; x = 2$

Table 3.1: Bit Assigned Table

ii) Borrow the Bits for Each Subnetwork:

Subnetwork	Range	
	Binary	Decimal

Cisco Network Lab	1010 1100. 0001 0010. 0000 0100. 0000 0000 - 1010 1100. 0001 0010. 0000 0100. 0011 1111	172.16.4.0 - 172.16.4.63
Embedded Lab	1010 1100. 0001 0010. 0000 0100. 0100 0000 - 1010 1100. 0001 0010. 0000 0100. 0111 1111	172.16.4.64 - 172.16.4.127
General Purpose Lab 1	1010 1100. 0001 0010. 0000 0100. 1000 0000 - 1010 1100. 0001 0010. 0000 0100. 1011 1111	172.16.4.128 - 172.16.4.191
General Purpose Lab 2	1010 1100. 0001 0010. 0000 0100. 1100 0000 - 1010 1100. 0001 0010. 0000 0100. 1111 1111	172.16.4.192 - 172.16.4.255
Core Switch	1010 1100. 0001 0010. 0000 0101. 0000 0000 - 1010 1100. 0001 0010. 0000 0101. 0000 1111	172.16.5.0 - 172.16.5.15
Access Point Ground Floor	1010 1100. 0001 0010. 0000 0101. 0001 0000 - 1010 1100. 0001 0010. 0000 0101. 0001 1111	172.16.5.16 - 172.16.5.31
Access Point First Floor	1010 1100. 0001 0010. 0000 0101. 0010 0000 - 1010 1100. 0001 0010. 0000 0101. 0010 1111	172.16.5.32 - 172.16.5.47
Video Conference Room	1010 1100. 0001 0010. 0000 0101. 0011 0000 - 1010 1100. 0001 0010. 0000 0101. 0011 1111	172.16.5.48 - 172.16.5.63
Hybrid Classroom	1010 1100. 0001 0010. 0000 0101. 0100 0000 - 1010 1100. 0001 0010. 0000 0101. 0100 1111	172.16.5.64 - 172.16.5.79
Lecturer Office	1010 1100. 0001 0010. 0000 0101. 0101 0000 - 1010 1100. 0001 0010. 0000 0101. 0101 1111	172.16.5.80 - 172.16.5.95
Printing Area and Small Meeting Room	1010 1100. 0001 0010. 0000 0101. 0110 0000 - 1010 1100. 0001 0010. 0000 0101. 0110 1111	172.16.5.96 - 172.16.5.111
Router	1010 1100. 0001 0010. 0000 0101. 0111 0000 - 1010 1100. 0001 0010. 0000 0101. 0111 0011	172.16.5.112 - 172.16.5.115

Table 3.2 : IP Address Range Table

3.1 Assign Address for Each Subnetwork

Subnetwork	Network Address	Broadcast Address	Usable Address Range	Subnet Mask
Cisco Network Lab	172.16.4.0	172.16.4.63	172.16.4.1 - 172.16.4.62	255.255.255.192
Embedded Lab	172.16.4.64	172.16.4.127	172.16.4.65 - 172.16.4.126	255.255.255.192

General Purpose Lab 1	172.16.4.128	172.16.4.191	172.16.4.129 - 172.16.4.190	255.255.255.192
General Purpose Lab 2	172.16.4.192	172.16.4.255	172.16.4.193 - 172.16.4.254	255.255.255.192
Core Switch	172.16.5.0	172.16.5.15	172.16.5.1 - 172.16.5.14	255.255.255.240
Access Point Ground Floor	172.16.5.16	172.16.5.31	172.16.5.17 - 172.16.5.30	255.255.255.240
Access Point First Floor	172.16.5.32	172.16.5.47	172.16.5.33 - 172.16.5.46	255.255.255.240
Video Conference Room	172.16.5.48	172.16.5.63	172.16.5.49 - 172.16.5.62	255.255.255.240
Hybrid Classroom	172.16.5.64	172.16.5.79	172.16.5.65 - 172.16.5.78	255.255.255.240
Lecturer Office	172.16.5.80	172.16.5.95	172.16.5.81 - 172.16.5.94	255.255.255.240
Printing Area and Small Meeting Room	172.16.5.96	172.16.5.111	172.16.5.97 - 172.16.5.110	255.255.255.240
Router	172.16.5.80	172.16.5.83	172.16.5.81 - 172.16.5.82	255.255.255.252

Table 3.1.1: IP Address Assigned Table

4.0 Virtual Local Area Network (VLAN)

To ensure network security and efficient traffic management, TTL Network Solutions has implemented a hierarchical VLAN strategy. By logically segmenting the physical network into distinct broadcast domains, we ensure that sensitive staff data is isolated from student labs and that the network remains scalable.

4.1 VLAN Table

The following table details the VLAN ID assignment, mapping each work area to its dedicated subnet.

VLAN ID	VLAN Name	Description	IP Subnet
10	VLAN_Cisco	Cisco Network Lab	172.16.4.0/26
20	VLAN_Embed	Embedded Lab	172.16.4.64/26
30	VLAN_GP1	General Purpose Lab 1	172.16.4.128/26
40	VLAN_GP2	General Purpose Lab 2	172.16.4.192/26
50	VLAN_Infra	Core Switch & WAPs	172.16.5.0/28
60	VLAN_VidConf	Video Conference Room	172.16.5.48/28
70	VLAN_Hybrid	Hybrid Classroom	172.16.5.64/28
80	VLAN_Staff	Lecturer Office	172.16.5.80/28
90	VLAN_Meet	Printing Area & Meeting Room	172.16.5.96/28

Table 4.1: VLAN Table

5.0 Range of IP Address Based on Different Area

5.1 Cisco Network Lab

VLAN ID & Name	
10, VLAN_Cisco	
IP Subnet	
172.16.4.0/26	
Default Gateway (SVI IP)	
172.16.4.1	
User	IP Address
Student and Lecturer (PCs)	172.16.4.8 - 172.16.4.62
Smart Board	172.16.4.2
Reserved for Future Growth	172.16.4.3 - 172.16.4.7

Table 5.1: Cisco Network Lab

5.2 Embedded Lab

VLAN ID & Name	
20, VLAN_Embed	
IP Subnet	
172.16.4.64/26	
Default Gateway (SVI IP)	
172.16.4.65	
User	IP Address
Student and Lecturer (PCs)	172.16.4.72 – 172.16.4.126
Smart Board	172.16.4.66
Reserved for Future Growth	172.16.4.67 – 172.16.4.71

Table 5.2: Embedded Lab

5.3 General Purpose Lab 1

VLAN ID & Name	
30, VLAN_GP1	
IP Subnet	
172.16.4.0 / 26	
Default Gateway (SVI IP)	
172.16.4.129	
User	IP Address
Student and Lecturer (PCs)	172.16.4.136 - 172.16.4.190
Smart Board	172.16.4.130
Reserved for Future Growth	172.16.4.131 - 172.16.4.135

Table 5.3: General Purpose Lab 1

5.4 General Purpose Lab 2

VLAN ID & Name	
40, VLAN_GP2	
IP Subnet	
172.16.4.192/26	
Default Gateway (SVI IP)	
172.16.4.193	
User	IP Address
Student and Lecturer (PCs)	172.16.4.200 – 172.16.4.254
Smart Board	172.16.4.194
Reserved for Future Growth	172.16.4.195 – 172.16.4.199

Table 5.4: General Purpose Lab 2

5.5 Video Conference Room

VLAN ID & Name	
60, VLAN_VidConf	
IP Subnet	
172.16.5.48/28	
Default Gateway (SVI IP)	
172.16.5.49	
User	IP Address
Conference Laptop	172.16.5.56 – 172.16.5.62
Smart TV / Display	172.16.5.50
Reserved for Future Growth	172.16.5.51 – 172.16.5.55

Table 5.5: Video Conference Room

5.6 Hybrid Classroom

VLAN ID & Name	
70, VLAN_Hybrid	
IP Subnet	
172.16.5.64/28	
Default Gateway (SVI IP)	
172.16.5.65	
User	IP Address
Student and Lecturer (PCs)	172.16.5.72 – 172.16.5.78
Smart Board	172.16.5.66
Reserved for Future Growth	172.16.5.67 – 172.16.5.71

Table 5.6: Hybrid Classroom

5.7 Lecturer Office

VLAN ID & Name	
80, VLAN_Staff	
IP Subnet	
172.16.5.0 / 28	
Default Gateway (SVI IP)	
172.16.5.81	
User	IP Address
Lecturer (PCs)	172.16.4.87 - 172.16.4.94
Reserved for Future Growth	172.16.4.82 - 172.16.4.86

Table 5.7: Lecturer Office

5.8 Printing Area and Small Meeting Room

VLAN ID & Name	
90, VLAN_Meet	
IP Subnet	
172.16.5.96/28	
Default Gateway (SVI IP)	
172.16.5.97	
User	IP Address
Network Printer / Laptops	172.16.5.104 – 172.16.5.110
Smart Board	172.16.5.98
Reserved for Future Growth	172.16.5.99 – 172.16.5.103

Table 5.8: Printing Area & Small Meeting Room

5.9 Access Point Ground Floor

VLAN ID & Name	
50, VLAN_Infra	
IP Subnet	
172.16.5.0 / 28	
Default Gateway (SVI IP)	
172.16.5.17	
User	IP Address
Wireless Access Point	172.16.5.18 - 172.16.5.23
Reserved for Future Growth	172.16.5.24 - 172.16.5.30

Table 5.9 : Access Point Ground Floor

5.10 Access Point First Floor

VLAN ID & Name	
50, VLAN_Infra	
IP Subnet	
172.16.5.32/28	
Default Gateway (SVI IP)	
172.16.5.33	
User	IP Address
Wireless Access Point	172.16.5.34 – 172.16.5.38
Reserved for Future Growth	172.16.5.39 - 172.16.5.46

Table 5.10 : Access Point First Floor

6.0 Conclusion

Task 5 of the project focused on designing and configuring an optimized IP addressing scheme within a structured academic network environment. By utilizing the assigned 172.16.4.0/23 network address, the project successfully implemented a subnetting design that ensures maximum scalability, logical division, and the optimal use of IP addresses. Every floor, laboratory, and functional space was meticulously designed to eliminate potential IP collisions and incorporate all necessary equipment, including workstations, patch panels, wireless access points, and projectors.

The rationale followed in implementing the subnetting design involved proper segmentation and the detailed allocation of subnet masks, IP ranges, and broadcast addresses for each specific device and area. To provide clear paths of communication between all devices, the addressing scheme focused on unique identification, logical organization, scalability, and security for future expansion.

Furthermore, as documented in our detailed meeting minutes, effective teamwork supported the cooperative effort required to achieve clarity in task functions and implementation. This design effectively achieves the current network load while simultaneously laying the essential groundwork for the institution's future network structure. Ultimately, this generalized and robust scheme ensures the network remains strong and effective, with the capacity to absorb any incoming traffic in the future.

7.0 Appendix

7.1 Member's Task / Work

Member	Task
Tey Xin Yi	● Identify network distribution for work areas of Ground Floor ● Assign address for each subnetwork ● Prepare VLAN table ● Identify range of IP address for areas of Ground Floor
Lee Hui Zhen	● Identify network distribution for work areas of First Floor ● Assign address for each subnetwork ● Prepare VLAN table ● Identify range of IP address for areas of First Floor
Tan Hui Shan	● Introduction ● Get Network Address from lecturer ● Conclusion ● Meeting Minute

7.2 Meeting Minute

DATE/TIME		22 December 2025 19:00	
LOCATION		Online(Google Meet)	
AGENDA		1. Understanding Task 5 requirements 2. Analysis Task 5 requirements 3. Confirm assigned network address from lecturer 4. Plain IP distribution for all work areas 5. Decide VLAN structure and network segmentation 6. Assign IP ranges for each lab and room 7. Task distribution	
Meeting MC		Lee Hui Zhen	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Tey Xin Yi		19:00	
Lee Hui Zhen		19:00	
Tan Hui Shan		19:00	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Understandin g Task 5 Requirements	- Tey explained Task 5 requirements, objectives, and marking rubric. - The group discussed the importance of IP addressing in avoiding IP conflicts, improving network performance, and supporting future expansion.	Tey (22/12)
2	Analysis of Task 5 requirements	Lee analyzed the requirement for detailed subnet calculations, VLAN usage, and structured IP tables.	Lee (22/12)
3	Network address confirmation	Tan confirmed the network address assigned by the lecturer is a /23 private network and will be used as the base network for subnetting.	Tan (22/12)

4	Network distribution planning	Tey suggested distributing IP addresses based on functional areas such as labs, classrooms, offices, and wireless zones to simplify management and reduce broadcast traffic.	Tey (22/12)
5	VLAN design decision	All members agreed to implement VLANs to separate lab networks, staff areas, and wireless access points for better security, performance, and traffic control.	All members (22/12)
6	IP range assignment	All members discussed and finalized IP ranges for Cisco Network Lab, Embedded Lab, General Purpose Labs, Video Conference Room, Hybrid Classroom, Lecturer Office, Printing Area, and access points on both floors.	All members (22/12)
7	Task Distribution	- Task distribution assigned as below: - Tey - Identify network distribution for work areas of Ground Floor - Assign address for each subnetwork - Prepare VLAN table - Identify range of IP address for areas of Ground Floor - Lee - Identify network distribution for work areas of First Floor - Assign address for each subnetwork - Prepare VLAN table - Identify range of IP address for areas of First Floor - Tan - Introduction - Get Network Address from lecturer - Conclusion - Meeting Minute	All members (22/12)
8	Meeting ended	21:00	

7.3 Rubric

IP Addressing:

- Use correct network address for group
- Workings is provided clearly and labelled
- IP division is appropriate and logical
- Complete detail of all IP assignation for all labs and room